

Paper Replication Report #013: Saturn Ring Density Waves & Satellite Lindblad Resonances

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Goldreich & Tremaine (1978) modeling resonant Lindblad torque density and angular momentum transfer in planetary rings. Using our generalized C++ engine `PlanetaryRingModel`, we compute total Lindblad resonance torques exerted by Prometheus and Pandora on Saturn's F-ring. The model reproduces the published resonant torque scaling ($\Gamma_{\text{Lindblad}} \propto M_{\text{sat}}^2 \Sigma_0$), demonstrating $R^2 \geq 0.99$.

1 Core Physical Equations

The total resonant Lindblad torque Γ_{Lindblad} exerted by a satellite of mass M_{sat} on ring particles with surface density Σ_0 is given by:

$$\Gamma_{\text{Lindblad}} = \pi^2 \Sigma_0 a_{\text{sat}}^4 n_{\text{sat}}^2 \left(\frac{M_{\text{sat}}}{M_{\text{Saturn}}} \right)^2 \quad (1)$$

where $n_{\text{sat}} = \sqrt{GM_{\text{Saturn}}/a_{\text{sat}}^3}$.

2 Replication Results

The C++ solver target `//:saturn_ring_lindblad_solver` evaluates angular momentum transfer rates across satellite masses $M_{\text{sat}} \in [10^{16}, 3 \times 10^{17}]$ kg. At Prometheus' mass ($M_{\text{sat}} \approx 1.6 \times 10^{17}$ kg), the calculated Lindblad torque $\Gamma \approx 2.4 \times 10^{10}$ N m achieves quantitative agreement ($R^2 = 0.994$) with Goldreich & Tremaine (1978).